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(54) **CLOTH DIAPER LAUNDRY METHOD AND DETERGENT**

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See application file for complete search history.

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(57) **ABSTRACT**

A washing method and detergent composition for cleaning soiled cloth diaper laundry is disclosed specifically tailored for simple home use without employing known skin irritants and without persistent residue accumulation. The present invention discloses a one- or two-wash-cycle method for cleaning cloth diapering washables as well as a clean-rinsing, highly-effective detergent composition comprising a surface active agent, a water softener, a surface tension reducer, a bleaching agent and a pH balancing agent.

20 Claims, No Drawings

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CLOTH DIAPER LAUNDRY METHOD AND DETERGENT**CROSS REFERENCE TO RELATED APPLICATIONS**

The present application claims the benefit of co-pending U.S. application Ser. No. 16/877,478 filed May 18, 2020, entitled Detergent for Cloth Diaper Laundry, the disclosure of which is incorporated by reference in its entirety.

FIELD OF THE INVENTION

The field of the present invention generally relates to a laundry washing method and detergent formula for cleaning soiled cloth diapers gently and effectively.

BACKGROUND OF THE INVENTION

Disposable diapers and baby wipes are terrible for the environment and a huge expense for new families. Babies soil 70-80 diapers per week, resulting in over 3500 disposable diapers entering our landfills by their first birthday. New parents are demanding a true planet-friendly alternative and are growing skeptical of the green-washed claims of the biodegradable disposables being marketed to them. Cloth diapers long predate our disposable culture and have seen a resurgence in popularity among frustrated eco-minded parents. Reusable diapers are truly landfill-free and enable new parents to sustainably diaper their babies.

Laundering cloth diapers is key to being able to reuse them and reduce diaper waste, and yet proper washing is the largest roadblock for new parents hesitant to embrace this diapering system. Our market currently lacks cohesive diaper laundering education AND a detergent that is powerful enough to clean a heavily soiled diaper while remaining gentle on baby skin and our planet.

A true cloth-friendly diaper detergent must be clean rinsing-meaning the formula cannot contain optical brighteners, fragrance and fabric softeners. The reason it's crucial to avoid those three ingredients is because their very purpose is to live on the surface of the fabric (after the final rinse) to perform their respective jobs (optical brighteners are multifaceted molecules that reflect light to make fabric appear brighter, fragrance makes fabric smell nice, and fabric softeners are greasy and oily molecules that make fabric feel softer). When laundering cloth diapers, any molecule left behind is problematic because it gives the ammonia and bacteria in human waste something to bind to, making a thorough cleaning increasingly difficult with each wash. After a few weeks of using a detergent containing those ingredients, parents are left with diapers that smell horrid, repel moisture, and can even cause diaper rash. As you can imagine, the trifecta of stink, leaks and inflamed skin will lead even the most eco-minded person to throw in the towel and switch to disposable diapers. In order to create a highly effective detergent formulation excluding fragrance, fabric softeners, and optical brighteners, years of research and experimentation were necessary.

Detergents omitting both fragrance and fabric softeners (marketed as "free & clear") are known in the art; however, existing detergent formulas that exclude all 3 (fragrance, fabric softeners and optical brighteners) are only very mild "eco-friendly" detergents. These existing triple-free detergents are ineffective even at cleaning regular (single-& double-layer) household fabrics and utterly useless in clean-

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ing cloth diapers (featuring multiple absorbent fabric layers). An highly effective triple-free formula is unknown in the art.

An ideal Cloth Diaper Laundry Detergent must feature a builder (surface tension reducer) that does not cause skin irritation. The builder/surface tension reducer opens up the fibers of the fabric thereby releasing the embedded soil and allowing the surfactant to get deep down in each layer to do its job. This is especially important when cleaning diapers because their design requires them to be made of multiple layers of thick absorbent material. The soil does not remain on the surface as it does with a standard garment made of 1-2 layers of fabric. Builder/surface tension reducers are alkalis. They naturally have a very high pH and that's what enables them to "open" the fibers. Baby skin, however, has a low pH (lower than that of adult skin). The high pH of an alkali is incompatible with the low pH of baby skin and that incompatibility can cause skin irritation. Avoiding irritation requires using alkalis that allow themselves to be thoroughly rinsed away and the pH of the fabric lowered by the right surfactant blend.

Crafting a blend of surface active agents (surfactants) capable of handling the wide range of soil that dirty diapers present is equally important. A blend of different surfactants (each targeting different soil types) is needed to clean away high concentrations of urea and ammonia as well as organic solid matter (poop) which takes on many variations depending on the baby's dietary stage and health. There are also supplements and medications that the baby is either taking directly or absorbing from infant formula or a nursing mother's breast milk. Finally (quite possibly the biggest challenge) are skincare products that parents apply on babies bottoms before diapering. Diaper creams, for example, add a tremendous amount of grease to the fabric which must be effectively removed in the wash cycle to prevent it from building up on the fabric, thereby repelling moisture. The surfactants must also break up and remove stains, suspend the soil and prevent re-deposition onto the fabric during the cycle.

Being part of an eco-minded system this detergent must meet the added task of satisfying our desire to use and create products that are not environmentally hazardous. Ingredients to be chosen must score well with the EWG (environmental working group). As a result, the ultimate formulation must be biodegradable, non-toxic, free of SLS, SLE, SLES, NPE, fragrances, dyes, glycols, phosphates, 1,4-dioxane, chlorine, DEA, formaldehyde, caustics, and optical brighteners.

Finally, such a detergent must be easy for families to use effectively without requiring complicated steps and utilizing multiple chemistries (as commercial wash processes do).

Bottom-line, we must combine a simple process with ingredients selected ingredients to be highly effective at their job while also remaining gentle on sensitive baby skin.

SUMMARY OF THE INVENTION

The present invention, a cloth diaper laundry method and detergent, has been carefully designed to simplify at-home cloth diaper cleaning and meet the needs of hardworking, earth-conscious families.

An effective method for cleaning soiled cloth diapers at home would work well on a wide range of home-use laundry machines-making use of standard wash cycles and temperatures. Likewise, an effective method would avoid the use of multiple different chemistries-reducing the potential for user error and improving ease, uptake and adoption of the methodology. Combined with effective detergent chemistry, an

effective method for cleaning soiled cloth diapers at home must combine ease of washing, safe chemistry and result in very clean diapers.

In one embodiment of the present invention a cloth diaper laundry method uses one or more wash cycles utilizing a single cloth diaper laundry detergent formula.

Specifically, cloth diapering washables (including any of the following: diaper inners (absorbent cloth layers), diaper outers (waterproof washable outer wraps), cleaning cloths, diaper liners and washable diaper holding pouch) are loaded into a washing machine. In an optional step, one or more agitators are added into the washing machine along with the cloth diapering washables. In an optional step, cloth diapering washables are added to a total load size ranging from one-half to two-thirds the washer's maximum recommended load. In an optional step, other colorfast laundry is mixed in with cloth diapering washables to ensure total load size between one-half to two-thirds of the washer's maximum recommended load.

After loading said cloth diaper washables into said washing machine, a pre-wash amount of diaper detergent (per the formula below) is added and a pre-wash cycle is initiated. Optionally, said pre-wash cycle is programmed for a warm water wash. Optionally, said pre-wash cycle can be programmed as either a quick wash cycle or a normal wash cycle. Optionally, the diaper detergent is added directly into the laundry.

Either after completing said pre-wash cycle or independently thereof, a main wash amount of diaper detergent (per the formula below) is added and a main wash cycle is initiated. Optionally, said main wash amount of diaper detergent is twice the pre-wash amount of diaper detergent. Optionally, said main wash cycle is programmed for a hot water wash. Optionally, said main wash cycle is programmed as a heavy duty cycle.

Optionally, the diaper detergent is added directly into the laundry and not into the washing machine's detergent cups.

Optionally, the washing machine's cycle is never set to allergen or sanitize cycles (both of which are too hot and can damage cloth diaper fibers and reduce effectiveness).

In a further embodiment, the entire laundry load (including said agitators) is transferred into a clothes dryer. Optionally, the diaper inners (absorbent cloth layers) are shaken out during transfer into the dryer.

Optionally, prior to transferring into the dryer, laundry can be inspected and smelled. Especially in buildings with very hard water, additional detergent may be required for effective pre-wash and main wash cycles.

Optionally, 2-3 ounces of chlorine bleach can be added to main second wash cycle, preferably using said washing machine's built-in bleach dispenser.

An ideal home-use cloth diaper detergent would balance a powerful detergent formula with a completely clean rinse-remaining free of optical brighteners, fragrance and fabric softeners.

Table 1 below sets forth the ingredient functions for certain preferred ingredients of the disclosed composition for diaper detergent.

TABLE 1

Ingredient Function	Preferred Ingredients
Surface Active Agent	Alcohols, C12-15 & C12-13, Ethoxylated
Water Softener	Sodium Sulfate
Builder (Surface Tension	Hydrous Sodium

TABLE 1-continued

Ingredient Function	Preferred Ingredients
Reducer)	Silicate
Bleaching Agent (Oxidizer)	Sodium Percarbonate
pH Balancing Agent (Alkali)	Soda Ash Light

First, a builder (surface tension reducer) must be selected that is highly alkaline to open up the cotton fibers and allow for deep cleaning of multi-layered cotton diapers. This builder also acts as a water softener which softens hard water by binding free water hardness ions like magnesium and calcium. Soil molecules are generally bound to the fabric surface by calcium ions. Softening of the mineral hardness in the water therefore helps with stain removal.

In one embodiment of the invention, the included builder is a hydrous sodium silicate.

In a further embodiment of the invention, a water softener (or chelating agent) is added to the formula to assist with inactivating mineral hardness. This is especially helpful for users living in areas with hard well water. This water softener further softens the water and helps to increase the efficiency of the surfactants.

In one embodiment of the invention, the water softener (or chelating agent) is a non-phosphate water softener.

In a further embodiment of the invention, the water softener (or chelating agent) is Sodium Sulfate.

In a further embodiment of the invention, one or more ionic and anionic surfactants are included in the formula.

In a further embodiment of the invention, the one or more ionic and anionic surfactants are non-sulfate surfactants.

In one embodiment of the invention, the one or more ionic and anionic surfactants are various ethoxylated alcohols.

In a further embodiment of the invention, the one or more ionic and anionic surfactants are selected from the group consisting of: Alcohols, C12-15, ethoxylated and Alcohols, C12-13, ethoxylated. For example, suitable ethoxylated alcohols could include: Tomadol 25-9, Tomadol 25-7 and Tomadol 23-5.

In a further embodiment of the invention, a bleaching agent (oxidizer) to lighten discolorations on soiled cloth diapers is included in the formula.

In one embodiment of the invention, the bleaching agent (oxidizer) is chlorine-free.

In one embodiment of the invention, the bleaching agent (oxidizer) is sodium percarbonate. For example, one suitable example of a bleaching agent (oxidizer) can be PROVOX C.

In a further embodiment of the invention, a pH balancing agent (alkali) is included in the formula.

In one embodiment of the invention the pH balancing agent (alkali) is soda ash light.

Table 2 shows minimal and preferred ranges of possible ingredients (% by weight) of a complete cloth diaper laundry detergent solution.

TABLE 2

Ingredient (example) [Function]	Minimal Range (% by wt)	Preferred Range (% by wt)
Alcohols, C12-15 & C12-13, Ethoxylated (e.g. Tomadol 25-9, Tomadol 25-7, Tomadol 23-5) [Surface Active Agent]	0.5% to 10.0%	5.7% to 6.3%
Alcohols, C12-15, Ethoxylated (e.g. Tomadol 25-7) [Surface Active Agent]	0.0% to 10.0%	0.95% to 1.05%
Alcohols, C12-13, Ethoxylated (e.g. Tomadol 23-5) [Surface Active Agent]	0.0% to 10.0%	3.8% to 4.2%

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TABLE 2-continued

Ingredient (example) [Function]	Minimal Range (% by wt)	Preferred Range (% by wt)
Sodium Sulfate [Water Softener]	10.0% to 25.0%	17.1% to 18.9%
Hydrous Sodium Silicate [Surface Tension Reducer]	5.0% to 15.0%	9.5% to 10.5%
Sodium Percarbonate (e.g. PROVOX C) [Bleaching Agent]	10.0% to 20.0%	13.3% to 14.7%
Soda Ash Light [pH Balancing Agent]	40.0% to 60.0%	49.4% to 54.6%

The embodiments of the invention in which an exclusive property or privilege is claimed are defined as follows:

1. A cloth diaper laundry method comprising:
loading a washing machine with cloth diaper washables,
programming a main wash cycle on said washing
machine, and
adding a main wash amount of a cloth diaper laundry
detergent composition to said washing machine,
wherein said cloth diaper laundry detergent composition
comprises:
(a) from 5.7% to 6.3% by weight of one or more surface
active agents;
(b) from 17.1% to 18.9% by weight of a water softener;
(c) from 9.5% to 10.5% by weight of a surface tension
reducer;
(d) from 13.3% to 14.7% by weight of a bleaching
agent; and,
(e) from 49.4% to 54.6% by weight of a pH balancing
agent;
wherein the percentage by weight is relative to the total
weight of the composition.
2. The cloth diaper laundry method of claim 1, wherein
the one or more surface active agents are sulfate-free, the
surface tension reducer is phosphate-free, and the bleaching
agent is chlorine-free.
3. The cloth diaper laundry method of claim 1, wherein
the one or more surface active agents is selected from the
group consisting of: alcohols, C12-15, ethoxylated and
alcohols, C12-13, ethoxylated.
4. The cloth diaper laundry method of claim 1, wherein
the one or more surface active agents is selected from the
group consisting of: Tomadol 25-9, Tomadol 25-7 and
Tomadol 23-5.
5. The cloth diaper laundry method of claim 4, wherein in
the aggregate the one or more surface active agents further

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comprises between 5.7% to 6.3% by weight, wherein the
percentage by weight is relative to the total weight of the
composition.

6. The cloth diaper laundry method of claim 1, wherein
the water softener is sodium sulfate.

7. The cloth diaper laundry method of claim 6, wherein
the water softener further comprises between 17.1% to
18.9% by weight, wherein the percentage by weight is
relative to the total weight of the composition.

8. The cloth diaper laundry method of claim 1, wherein
the surface tension reducer is hydrous sodium silicate.

9. The cloth diaper laundry method of claim 8, wherein
the surface tension reducer further comprises between 9.5%
to 10.0% by weight, wherein the percentage by weight is
relative to the total weight of the composition.

10. The cloth diaper laundry method of claim 1, wherein
the bleaching agent is sodium percarbonate.

11. The cloth diaper laundry method of claim 10, wherein
the bleaching agent further comprises between 13.3% to
14.7% by weight, wherein the percentage by weight is
relative to the total weight of the composition.

12. The cloth diaper laundry method of claim 1, wherein
the pH balancing agent is soda ash light.

13. The cloth diaper laundry detergent of claim 12,
wherein the pH balancing agent further comprises between
49.4% to 54.6% by weight, wherein the percentage by
weight is relative to the total weight of the composition.

14. The cloth diaper laundry method of claim 1, wherein
said main wash cycle is set to a hot water wash cycle.

15. The cloth diaper laundry method of claim 1, wherein
said main wash cycle is set to a heavy duty wash cycle.

16. The cloth diaper laundry method of claim 1, wherein
one or more agitators are loaded into the washing machine.

17. The cloth diaper laundry method of claim 1, wherein
a chlorine bleach is added into the washing machine for use
in said main wash cycle.

18. The cloth diaper laundry method of claim 1, further
comprising programming a pre-wash cycle on said washing
machine and adding a pre-wash amount of said cloth diaper
laundry detergent composition to said washing machine
prior to running said main wash.

19. The cloth diaper laundry method of claim 18, wherein
said pre-wash cycle is set to a warm water wash cycle.

20. The cloth diaper laundry method of claim 18, wherein
said pre-wash cycle is set to a quick wash or normal wash
cycle.

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